Combinatorics | Fall 2026 | MATH31003-01
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Homework 1

- **Due:** Thursday, August 27, 2026
- **Submit:** Gradescope, through Blackboard > Assignments > Homework
- **Topics:** The sum, product, difference, and division rules; permutations of strings

Exercises have the point values indicated below.

For each exercise:

- write your solution **by hand**;
- start each exercise on a new page and label the exercise clearly;
- explain your reasoning in words, supported by formulas and, when useful, figures and diagrams;
- justify each counting step and explain why your method counts the desired outcomes correctly;
- simplify your computation to a final numerical answer when the exercise asks for one;
- select the correct pages for that exercise when you upload your PDF to Gradescope.

Collaboration is encouraged, but write your solutions yourself. For each exercise, name anyone you worked with and any external tools you used. Do not use generative AI on these exercises.

Exercise 1: Using every letter

Points: 10

How many strings of length 10 over the alphabet $\{A, B, C\}$ contain each of the three letters at least once?

> **Need a free hint?**
>
> Count all strings, then subtract the forbidden strings. A forbidden string uses exactly one or exactly two different letters. How many use only A, only B, or only C? For each of the pairs $\{A, B\}$, $\{A, C\}$, and $\{B, C\}$, how many strings use both letters in the pair and no third letter?

Exercise 2: Forming a committee

Points: 10

A company has 20 employees in Marketing and 10 employees in Research and Development (R&D). It must form a committee with the following five positions:

- one chair;
- one vice-chair; and
- three ordinary members, whose positions are not distinguished from one another.

No employee may hold more than one position. The chair and vice-chair must come from different departments; the ordinary members may come from either department.

How many committees are possible?

> **Need a free hint?**
>
> Split into cases according to which department provides the chair and which provides the vice-chair. Choose the officers first, then the three ordinary members; remember that the ordinary-member positions are not distinguished.

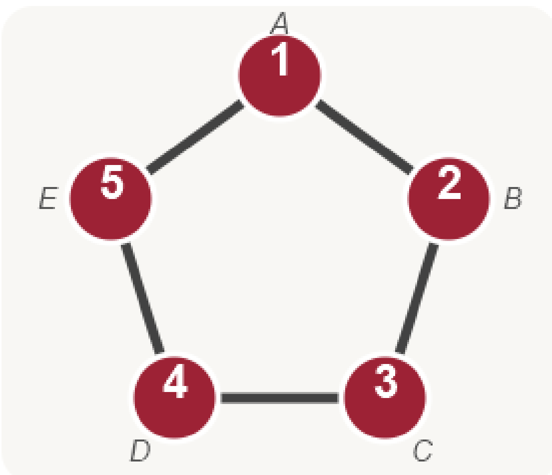
Exercise 3: Digits on a pentagon, without repetition

Points: 12

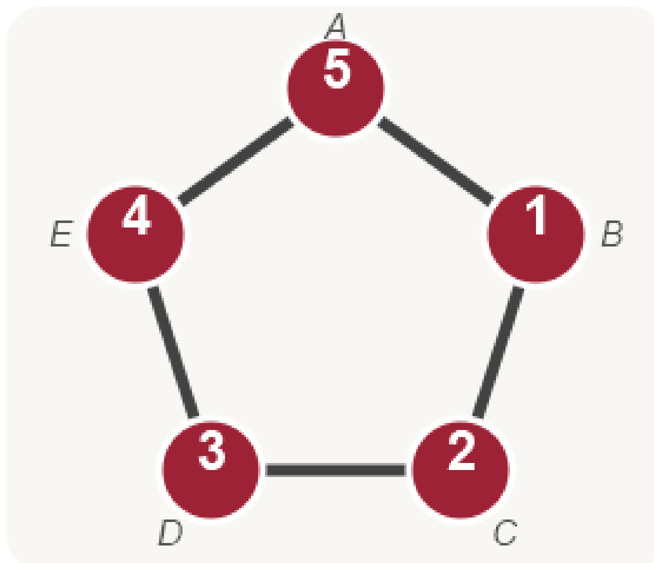
Place five **distinct** digits chosen from $0, 1, \ldots, 9$ at the vertices of a regular pentagon. Two patterns are considered the same when one can be obtained from the other by rotating the pentagon. Reflections are **not** considered the same.

The examples below illustrate this convention. The labels name the fixed vertex positions A, B, C, D, E, beginning at the top vertex and proceeding clockwise.

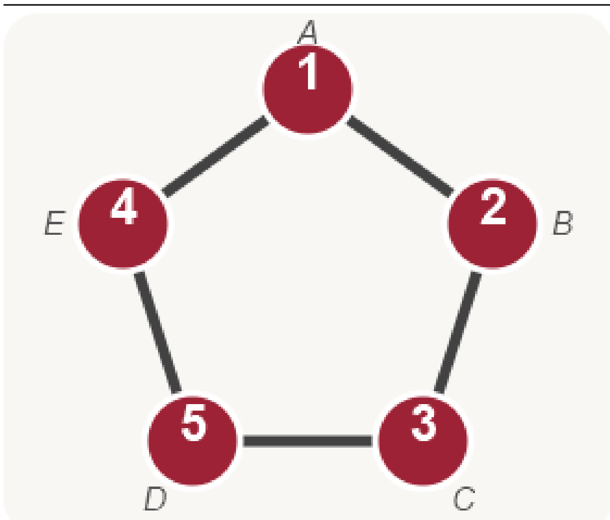
Pattern 1



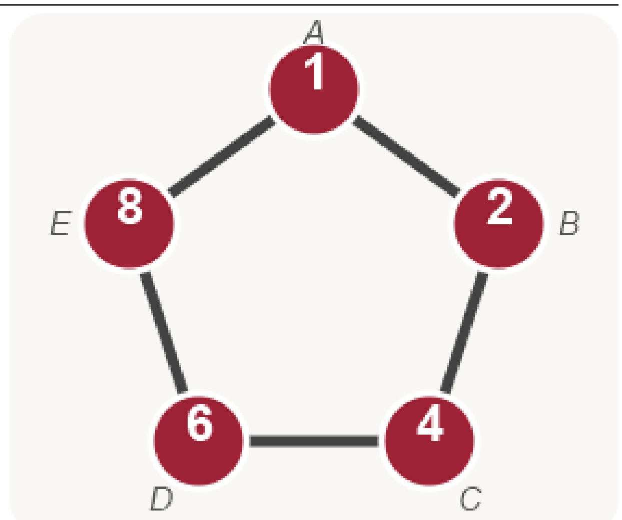
Pattern 2



Pattern 3



Pattern 4



Patterns 1 and 2 are considered the same because a rotation carries one to the other. No other pair among the four displayed patterns is considered the same.

How many different patterns are there?

Need a free hint?

First treat A, B, C, D, E as fixed, distinguishable positions and count the assignments of five distinct digits. Next determine how many labeled arrangements represent each pattern up to rotation, explaining why the "no repetition" condition makes this number the same for every pattern. Finally apply the division rule and simplify to a numerical answer.

Exercise 4: Why division by five can fail

Points: 8

Now change Exercise 3 by allowing digits to repeat. Thus each of the five vertices may receive any digit from $0, 1, \ldots, 9$, while patterns that differ only by a rotation are still considered the same.

A tempting argument is:

> There are 10^5 assignments to five fixed vertices, and every pattern is represented five times by rotation, so the answer is $10^5/5$.

Explain carefully why this argument is incorrect. You are **not** being asked to find the correct total.

> **Need a free hint?**
>
> Explain why 10^5 is the correct number of assignments when the vertices are fixed and labeled. Then list the distinct rotations represented by each of the clockwise strings `00000`, `00001`, and `01234`, beginning at vertex A, and use them to identify which requirement of the division rule fails. Focus on whether every pattern is counted the same number of times.